

Model Development Phase Template

Date	19 June 2025
Team ID	SWTID1749707061
Project Title	Online Payment Fraud Detection Using Machine Learning
Maximum Marks	5 Marks

Feature Selection Report:

In the forthcoming update, each feature will be accompanied by a brief description. Users will indicate whether it's selected or not, providing reasoning for their decision. This process will streamline decision-making and enhance transparency in feature selection.

Feature	Description	Selected (Yes/No)	Reasoning
Step	Time step (unit of time in the dataset)	Yes	Time patterns are often associated with fraudulent behaviour, especially if activities occur in off-hours or are spaced abnormally.
Type	Type of transaction (e.g., CASH_IN, TRANSFER, PAYMENT)	Yes	Certain transaction types (e.g., TRANSFER or CASH_OUT) have been observed to be more prone to fraud.
Amount	Amount of money involved in the transaction	Yes	Fraudulent transactions often involve unusually large or small amounts.

nameOrig	Identifier for the origin account	No	Being an ID, it holds no predictive power directly; better to encode using frequency or remove to prevent data leakage.
oldbalanceOrg	Account balance before the transaction (origin)	Yes	Comparing pre- and post-transaction balances can help detect inconsistencies or suspicious account behavior.
newbalanceOrig	Account balance after the transaction (origin)	Yes	Used in combination with <code>amount</code> and <code>oldbalanceOrg</code> to detect fraudulent or logically incorrect transactions.
oldbalanceDest	Account balance before the transaction (destination)	Yes	Helps assess whether the destination account behavior aligns with normal transaction flows.
newbalanceDest	Account balance after the transaction (destination)	Yes	Important for modeling legitimate fund flow and detecting anomalies.
isFraud	Target variable; whether the transaction is fraudulent	Yes	This is the classification target, used during training and evaluation.